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Predators like it hot: Thermal mismatch in a predator–prey system across an elevational tropical gradient

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Abstract

1. Climate change may have dramatic consequences for communities through both direct effects of peak temperatures upon individual species and through interspecific mismatches in thermal sensitivities of interacting organisms which mediate changes in interspecific interactions (i.e. predation). Despite this, there is a paucity of information on the patterns of spatial physiological sensitivity of interacting species (at both landscape and local scales) which could ultimately influence geographical variation in the effects of climate change on community processes. In order to assess where these impacts may occur, we first need to evaluate the spatial heterogeneity in the degree of mismatch in thermal tolerances between interacting organisms.
2. We quantify the magnitude of interspecific mismatch in maximum (CT_{max}) and minimum (CT_{min}) thermal tolerances among a predator–prey system of dragonfly and anuran larvae in tropical montane (242–3,631 m) and habitat (ponds and streams) gradients. To compare thermal mismatches between predator and prey, we coined the parameters maximum and minimum predatory tolerance margins (PTM_{max} and PTM_{min}), or difference in CT_{max} and CT_{min} of interacting organisms sampled across elevational and habitat gradients.
3. Our analyses revealed that: (a) predators exhibit higher heat tolerances than prey (~4°C), a trend which remained stable across habitats and elevations. In contrast, we found no differences in minimum thermal tolerances between these groups. (b) Maximum and minimum thermal tolerances of both predators and prey decreased with elevation, but only maximum thermal tolerance varied across habitats, with pond species exhibiting higher heat tolerance than stream species. (c) Pond-dwelling organisms from low elevations (0–1,500 m a.s.l.) may be more susceptible to direct effects of warming than their highland counterparts because their maximum thermal tolerances are only slightly higher than their exposed maximum environmental temperatures.

4. The greater relative thermal tolerance of dragonfly naiad predators may further increase the vulnerability of lowland tadpoles to warming due to potentially enhanced indirect effects of higher predation rates by more heat-tolerant dragonfly predators. However, further experimental work is required to establish the individual and population-level consequences of this thermal tolerance mismatch upon biotic interactions such as predator–prey.

KEYWORDS

biotic interactions, climate change, CT_{max}, CT_{min}, elevation, predation, PTM_{max}, warming tolerance

1 | INTRODUCTION

A primary goal in macrophysiology is to unveil variation in ecophysiological traits at large spatial scales and their ecological, evolutionary and conservation implications (Chown & Gaston, 2016; Gaston et al., 2009). At the species level, variation in critical thermal tolerances has demonstrated that geographic variation in the ability of organisms to tolerate high temperatures is less pronounced than tolerance to low temperatures (Sunday et al., 2019). Furthermore, direct impacts of increased temperatures appear to be higher in organisms distributed at low latitudinal and elevational areas because their warming tolerances (WT), an index of species´ ability to cope with changing conditions (i.e. difference between heat tolerance and maximum environmental temperature), are narrower (Deutsch et al., 2008; Sunday et al., 2014). At the community level, however, the field has some key challenges yet to be addressed. There is a paucity of information upon the patterns of large-scale physiological diversity of interacting species which could ultimately allow estimation of the direct (species level) and indirect (species interactions) cascading effects of climate change on community-level processes (Buckley, 2013). Integrating thermal tolerances of interacting organisms, such as predator and prey, with microclimatic data, will allow assessing how environmental conditions may be expected to inflict direct effects of warming on each species, by estimating independent WTs of predator and prey. However, the estimate of indirect effects of increased temperatures on biotic interactions within a physiological framework is generally lacking and focusing on species competitive interactions (Diamond et al., 2016, 2017). However, less information exists on indirect effects of predator–prey interactions (Cheng et al., 2017) and, to our knowledge, no examination of the spatial variation in the interactive pattern in any predator–prey system has been reported to date.

A first critical step towards the prediction of potential community-level impacts of climate warming is to examine the spatial variation in the degree of thermal tolerances mismatches between predator and prey communities. The process of predation in ectotherms (foraging, handling, killing, feeding and digestion) is highly temperature-dependent affecting both predators and prey (e.g. Dell et al., 2014; Vucic-Pestic et al., 2011). At warmer temperatures, predation rates of ectothermic organisms should increase with higher metabolic rates (Bennett, 1987). Empirical evidence suggests that some insect predators are adapted to warmer temperatures than both their prey and lower level consumers (Chown et al., 2015; Franken et al., 2018; Grigaltchik et al., 2012). At higher temperatures, predators possess higher attack velocities (Quenta Herrera et al., 2018), higher probabilities of capture (Anderson et al., 2001; Jara et al., 2019; Ohlund et al., 2015) and higher efficiency at handling prey (Sentis et al., 2013; Thompson, 1978). However, since warmer temperature increases organisms' growth rate (Anderson et al., 2001; Enriquez-Urzelai et al., 2019), it can also indirectly decrease predatory rates as prey size increases (Moore & Townsend, 1998; Tejedo, 1993). Conversely, there are situations where top predators may be less heat tolerant than their prey (Pincebourde et al., 2008; Twomey et al., 2012). In some systems, the carnivore trophic group appears to have peak performances at temperatures 10°C lower than herbivores (Dell et al., 2014) due to differences in activation energies (the life-dinner principle, Dawkins & Krebs, 1979; Dell et al., 2011, 2014), and thus, they are likely to be more sensitive to climatic alterations (Voigt et al., 2003). These previous publications suggest a thermal mismatch in thermal optima or CT_{max} between predators and consumers, analogous to that found in host–pathogen interactions (Cohen et al., 2017; Nowakowski et al., 2016). By exhibiting contrasting physiological thermal niches, the dynamic of predators and prey interactions may be altered by shifting temperatures. Mesocosm experiments have revealed significant indirect effects of warmer temperatures on the top-down effects of top predators and bottom-up forces in producers (Kratina et al., 2012; Shurin et al., 2012), that, ultimately, may lead to unexpected impacts at the community level that are likely to vary with local conditions (Urban et al., 2017). Communities distributed at habitats with high variation and peak temperatures (e.g. lowland temporary ponds, open areas) will likely show different physiological adaptations than those living in less variable habitats such as rivers or canopy-covered habitats (Kaspari et al., 2015; Pintanel et al., 2019). For instance, if predators have higher heat tolerances than prey, they may increase predation rates in warmer and more thermally variable sites (Harley, 2011), since organisms' performance is extremely reduced at temperatures close to heat tolerance (Angilletta et al., 2002). Given that peak temperature events are more common in ponds in contrast to streams, we predict that pond-dwelling predators should benefit more of broader interspecific mismatches in maximum thermal tolerances than their stream counterparts.

An ideal predator–prey system in which to test for a physiological thermal mismatch across large and local geographical scales is the dragonfly larvae (Odonata), as the predator, and anuran larvae (Amphibia), as prey. Dragonfly larvae are key predators for tadpoles in tropical and temperate fishless freshwater communities (Gascon & Travis, 1992; Van Buskirk, 1988). Recent studies in temperate areas suggest that dragonfly naiads may express higher heat tolerances than their aquatic prey (~6°C; Katzenberger et al., 2021). Eck et al. (2014) found that an increase in temperature decreased the survival of tadpoles of *Lithobates clamitans* exposed to two species of dragonfly naiad predators. Furthermore, damselflies are benefited from warmer temperatures by increasing predation (Wang et al., 2020) and growth rate (Van Dievel et al., 2017). Although the mechanism is unknown, this is thought to be due to an elevated encounter rate through increased prey activity in warmer temperatures, or through increased predator attack rates (Hayden et al., 2015).

By comparing the relative value of predator and prey CT_{max} and CT_{min}, we can evaluate contrasting thermal sensitivities between predators and prey that could be functionally relevant to assess the impact of a top consumer on the communities. In order to examine whether thermal sensitivities of predators and prey vary spatially across large-scale gradients (e.g. elevation) or between local habitats, we defined the maximum predatory tolerance margin (PTM_{max}), as the difference between maximum thermal tolerances (CT_{max}) of predator and prey, and, similarly, the PTM_{min}, as the differences of prey and predator CT_{min} (Figure 1a). These coefficients could be considered as a proxy of maximal and minimal thermal niche divergence of interacting species. Thus, a positive value of PTM_{max} or PTM_{min} indicates that predators would be capable to exploit niche spaces with more extreme peak temperatures than prey. In contrast, a negative value would indicate that prey are more thermotolerant than predators.

Here, we analyse the interspecific variation of critical thermal limits, critical thermal maximum (CT_{max}) and critical thermal minimum (CT_{min}) and the predatory thermal margins indexes (PTM_{max} and PTM_{min}), across a tropical mountain range within the predator–prey system of dragonfly and anuran larvae. We predict that: (a) a

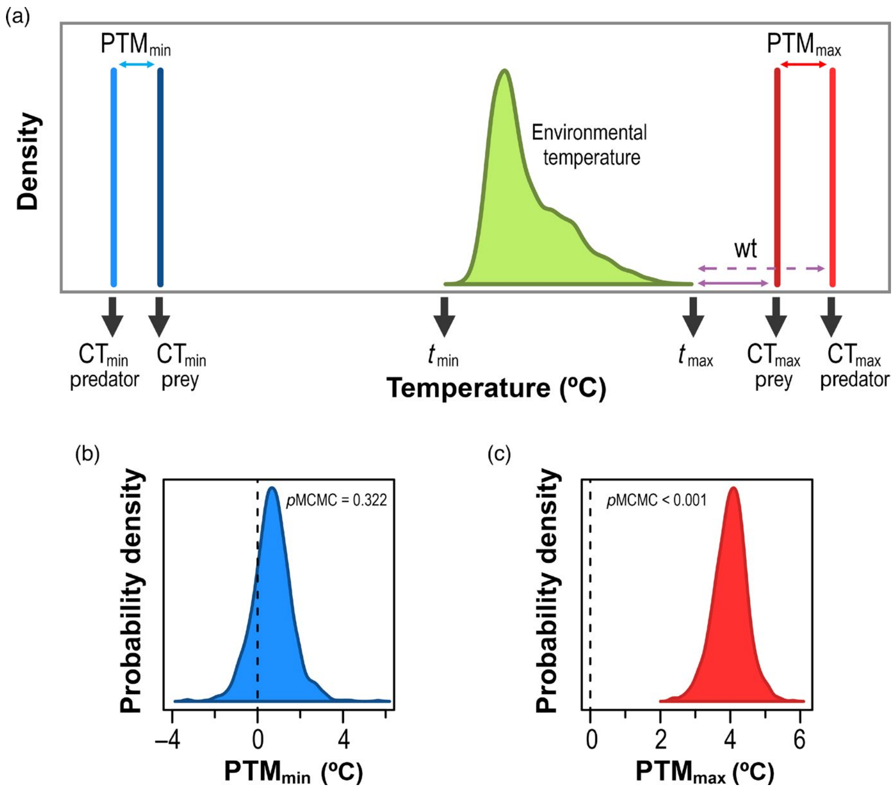

FIGURE 1 (a) Hypothetical freshwater predator–prey community. The critical thermal maximum (CT_{max}; red) and minimum (CT_{min}; blue) are the minimum and maximum body temperatures at which performance (i.e. movement) is possible for each species. Warming tolerance (WT; purple. Prey: solid line. Predator: dashed line) is the difference between species' CT_{max} and its current maximum environmental temperature (t_{max}; green). The maximum (PTM_{max}) and minimum (PTM_{min}) predatory tolerance margins are the difference between predator and prey maximum (CT_{max}) and minimum (CT_{min}) thermal tolerances respectively. (b and c) Predators have similar CT_{min} than prey but higher CT_{max} across 12 freshwater communities in an Ecuadorian mountain range. Frequency distribution of the intercept obtained from null models with (b) minimum and (c) maximum predatory tolerance margins as dependent variable

physiological mismatch exists between predator and prey's thermal tolerance, with dragonfly naiads being more heat and cold tolerant than their tadpole prey (positive PTM_{max} and PTM_{min}). (b) The degree of mismatch in thermal tolerance limits between trophic groups will differ between freshwater habitats (ponds and streams). We expect that predatory thermal margins should be positively correlated with environmental thermal variation, being wider in pond habitats than streams. Finally, (c) we expect that lowland pond species will be at higher risk from environmental warming than stream breeders or highland counterparts, because they would experience environmental temperatures closer to their maximum thermal tolerance, as previous studies suggested (e.g. Deutsch et al., 2008; Gutiérrez-Pesquera et al., 2016; Sunday et al., 2014).

2 | MATERIALS AND METHODS

2.1 | Study sites and thermal variability

We sampled predators and prey from 12 freshwater communities (seven ponds and five streams) in the Ecuadorian Andes (between latitudes 1°N and 3°S and elevations 242 and 3,631 m a.s.l.), from November 2014 to August 2016. Within each community, all individuals, both tadpole and naiad species, were collected from the same stream or pond, or nearby arrays of ponds, and transported to the laboratory (PUCE, Quito, Ecuador). We sampled the habitats randomly using a fish net. For one amphibian species (*Espadarana prosoblepon*) and given the impossibility of finding tadpoles, eggs were manually collected instead and maintained at laboratory conditions until they reached Gosner stage 25 (Gosner, 1960). A recent study suggests that critical thermal limits do not vary between previously acclimated captive-bred and field-caught tadpoles (Pintanel et al., 2020). Furthermore, the exclusion of *E. prosoblepon* from the analyses gave similar results. All species were housed separately at similar densities in plastic containers filled with 4 L of dechlorinated water. Full water changes were carried out three times a week. To allow acclimatisation to laboratory conditions, individuals were maintained at a constant temperature of 20°C using a portable thermal bath (patent license ES1135983U) with 12:12 light:dark for at least 3 days before the thermal tolerance measurements were conducted because in both insects and amphibians, thermal tolerances have been reported to stabilize after 2–3 days of exposure to a new thermal regimen (Allen et al., 2012; Brattstrom, 1968). Tadpoles were fed ad libitum rabbit chow. Each cage of predators received tadpoles ad libitum. We observed every species of dragonfly larvae feeding actively on tadpoles in captive conditions, so we assume that they are natural predators and prey in their native environments.

Each sampled community occupies different elevations and aquatic environments with contrasting temperatures. We characterised the thermal conditions within each habitat using temperature data loggers (HOBO UA-002-64) to obtain continuous records of habitat temperatures (every 15 min; see Table S1). At each collection site, data loggers were placed at the deepest bottom of the pond or stream, since it is the coolest region of the habitat and could be selected to avoid peak temperatures (Duarte et al., 2012). Given that water bodies are complex to categorize, we decided to use the two most restrictive levels of habitat type for the analyses, which are streams and ponds. For each data logger, we obtained three microclimatic thermal variables: minimum temperature (t_{min}), maximum temperature (t_{max}) and daily thermal range (dr).

2.2 | Thermal tolerance

We measured critical thermal tolerances of 465 tadpoles (232 CT_{max} and 233 CT_{min}) of 17 anuran species and 142 naiads (80 CT_{max} and 62 CT_{min}) of 20 dragonfly species (Table S2). However, given the limited number of dragonfly individuals for some species, we could only measure either CT_{max} (n = 19) or CT_{min} (n = 13). Since insects and amphibians tend to have reduced thermal tolerances during their metamorphic climax (e.g. Bowler & Terblanche, 2008; Enriquez-Urzelai et al., 2019; Floyd, 1983), we excluded dragonfly naiads near their last molt and anuran larvae over Gosner's stage 38 (Gosner, 1960). To obtain thermal tolerance estimates, we used the dynamic method of Lutterschmidt and Hutchison (1997a), which consists in increasing or decreasing the temperature at a constant rate (0.25°C/min) until an end point is attained. This end-point was defined as total immobility without any response to physical stimuli, which has the advantage of being reproducible for both thermal tolerance limits and taxa. To do so, each individual was placed in a 100-ml container with dechlorinated tap water in a thermal bath (HUBER K15-CC-NR Kältemaschinenbau AG) at a starting temperature of 20°C. Once the endpoint was attained, CT_{max} and CT_{min} were recorded as the temperature of the water next to the individual (Gutiérrez-Pesquera et al., 2016; Lutterschmidt & Hutchison, 1997b) with a quick-recording thermometer (Miller & Weber) to the nearest 0.1°C. Each individual was tested only once. After the test, the individuals were immediately transferred to plastic cups at the acclimation temperature (20°C) to allow for recovery. Only 2.7% of the dragonfly larvae and 0.8% of tadpoles died after the test. Given the low number of dragonfly naiad individuals, we included these data because they did not deviate from the ones that survived.

To evaluate the risk of each species suffering heat acute shocks, we estimated warming tolerance (WT; sensu Deutsch et al., 2008; Duarte et al., 2012) as the difference between heat tolerance and maximum environmental temperature ($WT = CT_{max} - t_{max}$). Finally, to describe the thermal tolerance mismatch between two interacting species, we calculated the predatory thermal margin as the difference between each predator and prey's CT_{max} ($PTM_{max} = CT_{max(predator)} - CT_{max(prey)}$) and CT_{min} ($PTM_{min} = CT_{min(prey)} - CT_{min(predator)}$) of those species that share sampling sites (Figure 1a).

2.3 | Phylogeny

We constructed the phylogenetic tree with the APE package in R (Paradis et al., 2004) using the most comprehensive amphibian (Jetz & Pyron, 2018) and odonate (Letsch et al., 2016) phylogenies to date. In some cases, dragonfly larvae could only be identified to the genus level; therefore, we included the species by randomly attaching branches on the genus representing the species. For some species of amphibians and genera of dragonflies not included in the phylogenies, we used the position of a known sister taxon that does appear in the phylogeny (Figure S1).

2.4 | Analyses

All our analyses, unless specified, were based on Bayesian phylogenetic mixed models (BPMMs) with Gaussian error structures, combined with Monte Carlo Markov chain approximations, as implemented in MCMCGLMM R-package (Hadfield, 2010). To test (a) if a physiological mismatch exists between predator and prey's thermal tolerances and (b) if the thermal mismatches differ between thermal habitats, we modelled the predatory thermal margins (PTM_{max} and PTM_{min}) as a function of two spatial macro (elevation) and micro (habitat: pond and streams) gradients. A significant intercept of the null model of predatory thermal margins would indicate that predators are either more (Intercept >0) or less (Intercept <0) thermotolerant than prey while an intercept not different to zero would indicate that predators and prey have similar critical thermal tolerances. To test (c) if the risk to suffer heat impacts changes with elevation and between habitats, we used the warming tolerance metric (WT) as dependent variable, with habitat and elevation as predictors. Given that both predatory thermal margins and warming tolerance are based on critical thermal limits, we finally examined how heat (CT_{max}) and cold (CT_{min}) tolerance vary with both, elevation and habitat. In all analyses, we ran multiple models including all possible combinations in order to maximize sample size either using the whole dataset or only within each trophic group. The phylogenetic structure was incorporated as a random effect for predatory thermal margins and for all within-trophic group critical thermal limits and warming tolerance analyses. For predatory thermal tolerance margins, we repeated the analyses twice using both phylogenetic trees containing the dragonfly or amphibian species and including the species of the other group not included in the phylogeny as random factor. For the analyses of critical thermal limits and warming tolerance including all species, we only included trophic group (tadpoles or dragonflies) as random effect. Because the physiological variables may vary across sites, we also modelled collection site as a random effect in all analyses. Given that estimates of thermal limits can be influenced by body size and predators are usually larger than prey, we analysed whether there was a relationship between size (i.e. log-transformed weight) and thermal limits. We found no relationship between weight and thermal limits, and therefore, weight was not included in subsequent analyses (Table S11; Figure S3). For each model, the Monte Carlo Markov chain was run for 1,010,000 iterations, with a 10,000 burn-in and a thinning interval of 1,000 iterations, resulting in 1,000 samples. For both the residual terms and the random effects, we used an inverse Wishart prior ($V = 1$, $v = 0.002$). We checked that autocorrelation levels among samples were lower than 0.1.

Freshwater habitats can be thermally complex, other factors rather than elevation and habitat may add further thermal variation throughout time and space (e.g. pond depth, shading, morphology, canopy cover). Thus, environmental temperatures should better explain physiological variation. We repeated the same previous procedure to test the relationship between physiological estimates (CT_{max}, CT_{min}, PTM_{max} and PTM_{min}) and environmental thermal variables (t_{max}, t_{min} and dr). Furthermore, temporal recording of temperatures varied across sites due logistics and unexpected events. However, given that annual temperature variation in the tropics is largely driven by diel thermal variation (Janzen, 1967; Shah et al., 2017; Figure S2), temporal variation in temperatures should be less important than differences between local microclimate. The use of in situ thermal data provides important information on local thermal variation (e.g. Pintanel et al., 2019; Table S3) that would be otherwise neglected with the commonly used macroclimatic temperature estimates. In order to exclude peak temperature events that could appear with longer periods of time, we repeated the analyses using mean t_{max} and mean t_{min}, which gave similar results (Tables S5 and S8). All the tested models are listed in Tables S4–S10. We report the posterior mean and the 95% credible intervals (CIs) for each variable, and assess significance according to *p*MCMC, which is the proportion of samples in the posterior distribution non-overlapping with zero and suggests whether the variables contribute to explaining variation in the response variable. All analyses were performed in R v3.6.1 (R Core Team, 2021).

3 | RESULTS

Dragonfly naiads were found to have higher CT_{max} estimates than anuran larvae in all sites (PTM_{max} Intercept (null model) = 3.96°C [2.87–4.79°C]; *p*MCMC < 0.001; Figures 1c and 2a). The highest maximum predatory tolerance margin was found in a partially canopy-covered pond at 1,066 m a.s.l. (PTM_{max} = 7.81°C) and the lowest in a stream at 1,200 m a.s.l. (PTM_{max} = 1.54°C). In contrast, we did not find differences in CT_{min} between dragonflies and tadpoles (PTM_{min} Intercept [null model] = 0.75°C [−0.75 to 4.48°C]; *p*MCMC = 0.322; Figures 1b and 2b).

We found no difference in PTM_{max} and PTM_{min} with elevation or across habitats (Figure 3c,d; Table S4) nor with daily temperature variation or peak temperatures (*p*MCMC ≥ 0.324; Table S5). However, organisms inhabiting streams presented lower CT_{max} than their pond-dwelling conspecifics (Figures 2a and 3a) but similar CT_{min} (Figures 2b and 3b). Both CT_{max} and CT_{min} declined with elevation being steeper the decrease in CT_{min} (Figure 3a,b; Table S6 and S7). Furthermore, we found a positive relationship between CT_{max} and t_{max} ($ß = 0.208$ [0.106–0.321]; *p*MCMC = 0.002; Table S8) and

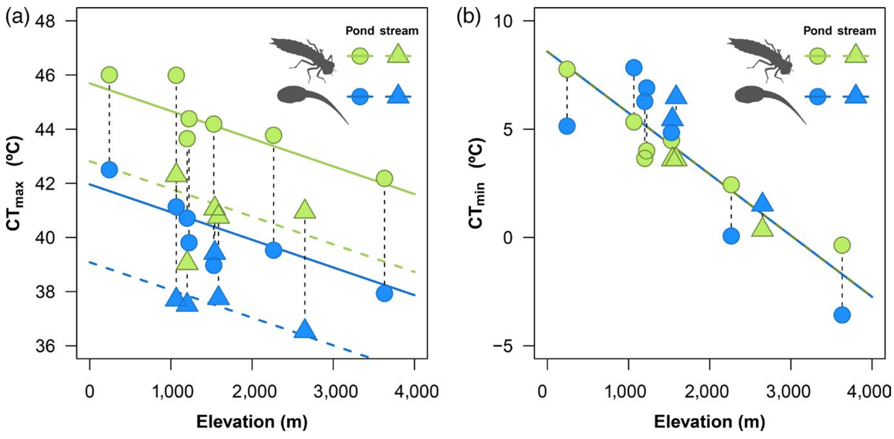

FIGURE 2 Variation in CT_{max} (a) and CT_{min} (b) in dragonfly larvae (green) and tadpoles (blue) along a tropical elevational range in Ecuador and across microhabitats (pond: circle and stream: triangle). For visual purposes, each value represents the mean CT_{max} or CT_{min} for each taxa within a locality. The black vertical dashed line connects predator and prey (a) CT_{max} and (b) CT_{min} in each locality and their extent define the thermal mismatch between predator and prey, in either CT_{max} (PTM_{max}) or CT_{min} (PTM_{min}) respectively. The fitted lines in the figure are derived from the raw data

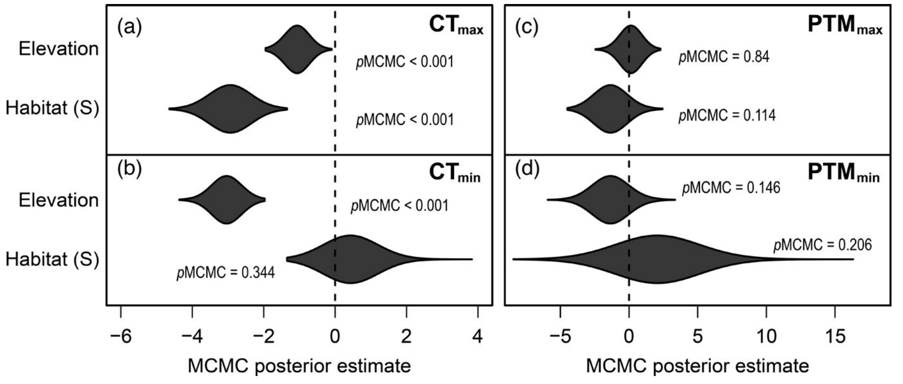

FIGURE 3 Distribution of coefficient estimates of models predicting (a) critical thermal maximum (CT_{max}), (b) critical thermal minimum (CT_{min}), (c) maximum predatory thermal margin (PTM_{max}) and (d) minimum predatory thermal margin (PTM_{min}) considering elevation (in km) and habitat (pond or S:stream) as predictors (for details, see Tables S2 and S5)

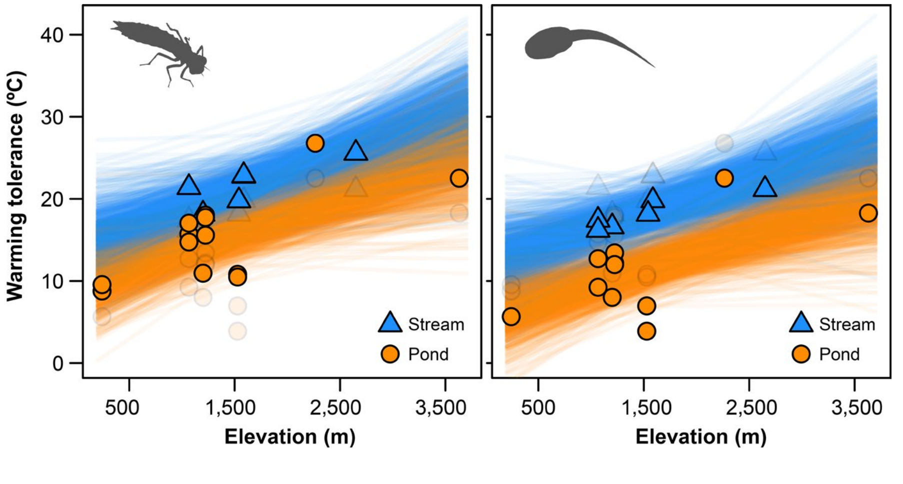

FIGURE 4 Variation in warming tolerance, WT ($WT = CT_{max} - t_{max}$) in dragonfly larvae (left panel) and tadpoles (right panel) along a tropical elevational range in Ecuador and across microhabitats (stream: triangle and pond: circle; for details, see Tables S9 and S10). In both panels, transparent points projected the alternative, either tadpole (left panel) or dragonfly (right panel), data for comparative purposes. See that all species within each freshwater habitat were included

between CT_{min} and t_{min} ($ß = 0.496$ [0.276–0.731]; $pMCMC < 0.001$; Table S8).

Warming tolerances reached a minimum at lower elevations (0–1,500 m; $ß = 0.004$ [0.002–0.007]; $pMCMC = 0.004$) and in ponds as opposite to streams ($ß = 5.836$ [1.083–10.68]; $pMCMC = 0.03$; Figure 4; Table S8). Given that dragonfly larvae have higher heat tolerances than tadpoles (Figures 1c nad 2a); tadpoles inhabiting lowland ponds are expected to be more vulnerable to warming since maximum temperatures are closer to their CT_{max} (i.e. smaller warming tolerance).

4 | DISCUSSION

To assess the potential ecological impacts of climate change, understanding the spatial patterns of physiological diversity across interacting species remains one of the greatest challenges for macrophysiology (Chown & Gaston, 2016). In this study, we demonstrate that predatory dragonfly larvae have higher maximum thermal tolerances than their tadpole prey (Figure 1c). We did not find support for our prediction of increased thermal mismatch with more variable environments. Instead, the results support the conclusion that dragonfly predators tolerate higher maximum peak temperatures than anuran prey throughout a broad tropical elevational range (3,400 m), and both lentic and lotic freshwater habitats. Predators have a sustained higher thermal resistance than prey across environmental gradients, even in the cooler streams where both tadpoles and dragonflies showed warming tolerances above 10°C (Figure 4). Although our approach has provided, to our knowledge, the most extensive analyses of thermal tolerances within a predator–prey system, we cannot rule out the possibility that the relationship between predatory thermal margins and environmental thermal variability could arise if more field sites are included. Previous studies with other terrestrial and marine predator–prey systems have shown contrasting results with prey presenting either higher or lower CT_{max} than their predators (Cheng et al., 2017; Coombs & Bale, 2014; Franken et al., 2018; Hughes et al., 2010; de Mira-Mendes et al., 2019; Monaco et al., 2016; Stoks et al., 2017). Those differences may be idiosyncratic and related to specific predation and evasion mechanisms within each system and therefore should be analysed independently.

We found that heat tolerance is more variable locally and across taxa, while cold tolerance seems to be more variable throughout large geographic climatic and elevational gradients (Muñoz & Bodensteiner, 2019; Muñoz et al., 2014; Pintanel et al., 2019). This mismatch in thermal tolerance has numerous implications for the studied predator–prey system under climate change. By exhibiting higher heat tolerances than their prey, dragonfly naiads should maintain physiological performance at a higher temperature than their prey. This may allow them to take advantage of episodes of rising water temperature (Eck et al., 2014; Richards & Bull, 1990; but see Smolinský & Gvoždík, 2014), such as those occurring during pond desiccation (Reques & Tejedo, 1995). This relationship between thermal mismatch and predation success has been observed in an analogous insect and tadpole predator–prey system. de Mira-Mendes et al. (2019) showed that invertebrate predators, which possess higher thermal tolerance than their tadpole prey, increase their predation rate at higher temperatures. Additionally, damselflies increase food intake and ultimately growth rate with heat waves (Van Dievel et al., 2017). This suggests that thermal mismatches between predator and prey may benefit the predators at warmer conditions. However, responses of organisms to warming may differ with short-term and longer term exposure to stressful temperatures (Rezende et al., 2020). For instance, the increase of the metabolic costs of the predators (and also prey) with long-term exposure to high temperatures may reduce performance and raise mortality (Gunderson & Leal, 2016; but see Van Dievel et al., 2017) even if predators are able to catch prey more easily. Furthermore, if the abundance of prey is reduced, survival of predators may be compromised by lack of food or through an increase of intraguild predation. Future work should focus on analysing the relationship between predatory tolerance margins and predation rates across different temperatures including realistic experimental approaches under natural or semi-natural conditions (e.g. mesocosms).

We used a conservative approach for the mismatch analyses since we gave the same priority to all predator and prey. In extreme weather events, it is likely that only the most thermally tolerant predators within a freshwater environment may survive or benefit from increasing temperatures. Thus, using only the most thermotolerant predator species might give a better picture of the outcomes in extreme thermal events. Furthermore, warmer temperatures may promote predator or prey replacements by more thermotolerant taxa, altering the current previsions. For instance, Amundrud and Srivastava (2020) found that, under drought, insect distribution in tropical mountains was mediated by species interactions rather than physiological effects of environmental conditions. Thus, warming temperatures may have unpredictable outcomes on complex community-level responses. Further studies across different global ecoregions may shed further light upon whether these findings are representative of freshwater predator–prey interactions globally.

Lowland tropical and subtropical anurans are already experiencing temperatures close to their CT_{max} (Figure 4; Duarte et al., 2012; von May et al., 2019). As habitats increase in temperature, the future challenge facing pond-dwelling lowland tadpoles will be to avoid both direct thermal stress and also to counter the potential indirect effects of enhanced predation rates by higher heat tolerant predators. Tadpoles will eventually encounter temperatures reaching or surpassing their maximum thermal tolerances, which is likely to challenge their survival. Analogous to other ectotherms, amphibians may respond to increasing temperatures by shifting to cooler environments. This can be achieved through migration to higher elevation sites (Lenoir & Svenning, 2013; Raxworthy et al., 2008) or modifying behaviour to use cooler microenvironments (e.g. from open to canopy-covered ponds). Organisms may further respond in situ by shifting their physiology to adapt to these higher environmental temperatures. Behavioural and acclimation responses might be important for buffering initial increase of temperatures (Enriquez-Urzelai et al., 2020), but later evolutionary responses will be needed for populations resilience (Buckley & Huey, 2016; Geerts et al., 2015).

In summary, our results show that within this predator–prey system, dragonfly naiad species exhibit consistently higher heat tolerance than tadpoles throughout a broad elevational range (~3,400 m) and across habitats. Thus, tadpoles should be more vulnerable to global warming than their dragonfly larvae predators. Especially vulnerable should be those occupying ponds at low elevations (<1,500 m), as a result of their heat tolerance being only slightly higher than the environmental temperatures they are currently exposed. Furthermore, we propose that this consistent mismatch in heat tolerances may have indirect effects on predation pressure at peak temperatures, potentially increasing the vulnerability of these tadpole communities. This could drive changes in amphibian distribution ranges in tropical mountains with possible biotic attrition or loss of local diversity in lowland tropical mountain communities (Colwell et al., 2008; Feeley & Silman, 2010). Such patterns of variation in physiological tolerance suggest that future warming is likely to heterogeneously influence biotic interactions at a landscape scale.

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AUTHORS' CONTRIBUTIONS

P.P., M.T. and A.M.-V. designed the study; P.P., M.T. and S.S.-I. collected the individuals; P.P. performed the experiments; P.P. analysed the data; all authors contributed to the writing.

DATA AVAILABILITY STATEMENT

Data available from the Dryad Digital Repository https://doi.org/10.5061/dryad.g79cnp5ps (Pintanel et al., 2021).

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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